

MAINFRAME DISCOVERY PLATFORM IMPLEMENTATION GUIDE

Objective

Build a reusable Mainframe Discovery Platform capable of analyzing large enterprise mainframe repositories containing:

- COBOL
- JCL
- Copybooks
- DB2
- CICS
- Control-M/Scheduler artifacts

The platform should support:

- Repository discovery
- Dependency analysis
- Root program discovery
- Data lineage
- Impact analysis
- Business rule extraction
- Natural language Q&A
- Future modernization

Workspace Structure

mainframe-discovery/

|—— mfcode/ | |—— discovery/ | |—— backend/ | |—— parsers/ | |—— graph/ | |——
knowledge/ | |—— docs/ | |—— tests/ | |—— .github/ | |—— .skills/

STEP 1

Create Global Copilot Instructions

File:

.github/copilot-instructions.md

Contents:

You are a Principal Mainframe Modernization Architect.

Expertise:

- COBOL
- JCL
- DB2
- CICS
- T-SYS
- Enterprise Batch Systems
- Knowledge Graphs
- Python 3.13
- FastAPI
- Pydantic v2
- Neo4j
- NetworkX

Rules:

1. Architecture first.
2. Explain design decisions.
3. Use production-grade patterns.
4. Design for very large repositories.
5. Generate tests.
6. Generate logging.
7. Generate error handling.
8. Prefer modular architecture.
9. Design for future AI integration.
10. Avoid throwaway code.

Required Output Format:

1. Problem Analysis
2. Architecture
3. Domain Models
4. Implementation Plan
5. Code
6. Tests
7. Future Improvements

STEP 2

Create Repository Discovery Skill

File:

.skills/repository-discovery/skill.md

Purpose:

Understand repository structure.

Responsibilities:

- Identify COBOL
- Identify JCL
- Identify Copybooks
- Identify DB2 artifacts
- Identify CICS artifacts

Output:

Repository Inventory

Prompt:

Act as a Mainframe Discovery Engineer.

Analyze repository structure.

Identify:

- COBOL
- JCL
- COPYBOOK
- DB2
- CICS

Generate:

1. Repository inventory strategy
2. Metadata schema
3. Extraction workflow
4. Scalability considerations

Do not generate implementation yet.

STEP 3

Create COBOL Analysis Skill

File:

.skills/cobol-analysis/skill.md

Purpose:

Analyze COBOL metadata.

Extract:

- PROGRAM-ID
- CALL
- COPY
- EXEC SQL
- FILE definitions

Output:

Program Catalog

Prompt:

Act as a COBOL compiler engineer.

Design a scalable COBOL parser.

Extract:

- PROGRAM-ID
- CALL statements
- COPY statements
- EXEC SQL
- FILE definitions

Requirements:

- Incremental parsing
- Large repository support
- Testability
- Pydantic models

Output:

1. Architecture
2. Domain Models
3. Parsing Strategy
4. Edge Cases

Do not generate code yet.

STEP 4

Create JCL Analysis Skill

File:

.skills/jcl-analysis/skill.md

Purpose:

Identify batch entry points.

Extract:

- JOB
- PROC
- EXEC PGM
- DD statements

Output:

Job Catalog

Prompt:

Act as a Senior Mainframe Batch Architect.

Design a JCL parser.

Extract:

- JOB
- PROC
- EXEC PGM
- DD
- Dataset references

Create relationships:

JOB → PROGRAM

Output:

1. Architecture
 2. Domain Models
 3. Parsing Strategy
 4. Edge Cases
-

STEP 5

Create DB2 Lineage Skill

File:

.skills/db2-lineage/skill.md

Purpose:

Create data lineage.

Extract:

- SELECT
- INSERT
- UPDATE
- DELETE

Output:

Program → Table relationships

Prompt:

Act as a DB2 lineage specialist.

Design a DB2 metadata extraction service.

Extract:

- Operations
- Tables
- Columns

Output:

1. Metadata Model
 2. Extraction Strategy
 3. Lineage Design
 4. Scalability Approach
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STEP 6

Create Graph Builder Skill

File:

.skills/graph-builder/skill.md

Purpose:

Build dependency graph.

Nodes:

- Program
- Job
- Table
- Copybook
- Dataset

Edges:

- CALLS
- EXECUTES
- USES
- READS
- WRITES
- UPDATES

Prompt:

Act as a Knowledge Graph Architect.

Design a dependency graph.

Support:

- Impact Analysis
- Root Discovery
- Lineage
- Traceability

Output:

1. Graph Model
2. Node Types
3. Edge Types
4. Query Patterns

STEP 7

Create Impact Analysis Skill

File:

.skills/impact-analysis/skill.md

Purpose:

Determine downstream impacts.

Prompt:

Act as an Enterprise Impact Analyst.

Design impact analysis workflows.

Questions:

- What breaks if table changes?
- What breaks if copybook changes?
- What breaks if program changes?

Generate:

1. Impact Model
 2. Traversal Strategy
 3. Query Patterns
-

STEP 8

Create Modernization Advisor Skill

File:

.skills/modernization-advisor/skill.md

Purpose:

Prepare future modernization.

Prompt:

Act as a Mainframe Modernization Architect.

Using dependency graph and lineage:

Generate:

- Service candidates
- Bounded contexts
- API candidates
- Event candidates
- Batch decomposition opportunities

Output architecture only.

IMPLEMENTATION ORDER

Phase 1

Repository Discovery

Deliverable:

Inventory

Phase 2

COBOL Analysis

Deliverable:

Program Catalog

Phase 3

JCL Analysis

Deliverable:

Job Catalog

Phase 4

DB2 Analysis

Deliverable:

Lineage Catalog

Phase 5

Graph Builder

Deliverable:

Knowledge Graph

Phase 6

Impact Analysis

Deliverable:

Dependency Explorer

Phase 7

Natural Language Q&A

Deliverable:

Mainframe Discovery Assistant

Questions:

- Which jobs execute program X?
 - Which programs call program X?
 - Which tables are updated by program X?
 - Show root program.
 - Trace transaction flow.
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COPILLOT USAGE RULE

Always use this sequence:

1. Ask for Architecture
2. Ask for Review
3. Ask for Improvements
4. Ask for Implementation
5. Ask for Tests
6. Ask for Code Review

7. Ask for Refactoring

Never start with:

"Write the code"

Always start with:

"Design the architecture"

This single habit will dramatically improve Copilot output quality.